Hints: Limits – 2. Recurrences solvable in closed form

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. By induction $x_n > 0$. Compute $x_n - \sqrt{c}$ and $x_n + \sqrt{c}$ separately: $x_n \pm \sqrt{c} = \dfrac{(x_{n-1} \pm \sqrt{c})^2}{2x_{n-1}}$. Hence the ratio $q_n = \dfrac{x_n - \sqrt{c}}{x_n + \sqrt{c}}$ satisfies $q_n = q_{n-1}^2$. (This is the substitution $x_n = \sqrt{c}\coth\theta_n$, which turns the recurrence into $\theta_n = 2\theta_{n-1}$.)

Hint 2. So $q_n = q_0^{2^n}$, where $|q_0| < 1$ because $x_0 > 0$, and $x_n = \sqrt{c}\,\dfrac{1+q_n}{1-q_n} \to \sqrt{c}$. Moreover $x_n - \sqrt{c} = \dfrac{2\sqrt{c}\,q_0^{2^n}}{1 - q_0^{2^n}}$: for $n \geqslant 1$ the error is nonnegative and is essentially squared at each step (quadratic convergence), so the number of correct digits roughly doubles each step.

Hint 3. Newton's method for $g(x) = x^2 - c$ is $x_n = x_{n-1} - g(x_{n-1})/g'(x_{n-1})$; check that this is exactly the given recurrence.

Problem 2

Hint 1. Following the suggestion, express $a_n = (b_n^2 - 1)/24$ and compute $b_{n+1}^2 = 1 + 24a_{n+1}$ in terms of b_n. It turns out to be a perfect square.

Hint 2. One gets $b_{n+1}^2 = \left(\frac{b_n+3}{2}\right)^2$, so $b_{n+1} = \frac{b_n+3}{2}$ with $b_1 = 5$, a linear recurrence: $b_n - 3 = 2^{2-n}$.

Hint 3. Hence $a_n = \dfrac{(3 + 2^{2-n})^2 - 1}{24} = \dfrac{1}{3} + \dfrac{1}{2^n} + \dfrac{2}{3 \cdot 4^n} = \dfrac{(2^n+1)(2^n+2)}{3 \cdot 4^n}$.

Problem 3

Hint 1. Hyperbolic substitution in algebraic form: write $a_n = \dfrac{u_n + u_n^{-1}}{\sqrt{5}}$ with $u_n \geqslant 1$ (i.e. $a_n = \frac{2}{\sqrt{5}}\cosh\theta_n$). Then $5a_n^2 - 4 = (u_n - u_n^{-1})^2$, so the root equals $u_n - u_n^{-1}$.

Hint 2. With $\tau = \frac{1+\sqrt{5}}{2}$ one has $\frac{3\pm\sqrt{5}}{2} = \tau^{\pm 2}$, and the recurrence becomes $a_{n+1} = \dfrac{\tau^2 u_n + \tau^{-2}u_n^{-1}}{\sqrt{5}}$, i.e. $u_{n+1} = \tau^2 u_n$. Since $\tau + \tau^{-1} = \sqrt{5}$, $u_0 = \tau$ and $u_n = \tau^{2n+1}$; by Binet's formula (odd index) $a_n = \dfrac{\tau^{2n+1} + \tau^{-(2n+1)}}{\sqrt{5}} = F_{2n+1}$, the Fibonacci numbers.

Hint 3. Compute $b_n = 0, 1, 1, 4, 9, 25, \ldots$ and prove $b_n = F_n^2$ by induction: $b_{n+1} = a_n - b_n = F_{2n+1} - F_n^2 = F_{n+1}^2$ by the identity $F_n^2 + F_{n+1}^2 = F_{2n+1}$. Finally $b_{2n+1} = F_{2n+1}^2 = a_n^2$.

Problem 4

Hint 1. Theorem 1: the characteristic roots of $\lambda^2 - 6\lambda + 8$ are 2 and 4; the right-hand side $n = n \cdot 1^n$ and 1 is not a root, so look for a particular solution $An + B$.

Hint 2. Substituting gives $A = \frac{1}{3}$, $B = \frac{10}{9}$. Then fit $a_n = C2^n + D4^n + \frac{n}{3} + \frac{10}{9}$ to $a_0 = 1$, $a_1 = 2$.

Hint 3. $C = -\frac{1}{2}$, $D = \frac{7}{18}$, i.e. $a_n = \dfrac{7 \cdot 4^n - 9 \cdot 2^n + 6n + 20}{18}$ (check: $a_2 = 6$).

Problem 5

Hint 1. Write $\alpha = \cos\theta$ with $\theta \in [0, \pi]$ and show by induction that $a_n = \cos n\theta$, using $\cos(n+2)\theta + \cos n\theta = 2\cos\theta\cos(n+1)\theta$. Then $|a_n| \leqslant 1$ is immediate.